PokeBench: Reproducible Decision-Making Benchmarks in Pokemon Red

afk-sapien

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Abstract PokeBench compares language-model agents on checkpoint tasks in Pokemon Red. Tasks cover navigation, battle tactics, manual menu operation, team preparation, and quests. Structured player information and mechanical shortcuts support strategy tasks, while menu tasks require manual inputs. This report presents a completed cross-provider pass, its token usage, a review of failure traces, and a targeted repeatability check. Opus obtained the highest equal-category score in the initial pass, while category results exposed different strengths across models. Many tasks reached a ceiling or floor, and response-format failures clustered in the warp-navigation task. The results describe agents within this framework, not model ability independent of tools. All original attempts remain recorded, including repaired infrastructure failures. Repeat outcomes are reported separately so that selected tasks do not receive extra weight in the main ranking.

1. Introduction

Playing Pokemon requires navigation, resource management, menu operation, battle tactics, and quest completion. A full campaign tests these skills together, but early navigation failure can hide later battle ability. PokeBench uses checkpoints to test them separately. PokeBench Long measures progress through a full campaign and is outside this report.

The focused pass covered 48 tasks and 50 registered starts per model. It produced 300 finished attempts across 6 models. The analysis retains useful results through routine infrastructure repairs. It records changed conditions and archived charges rather than discarding the earlier evidence.

The measured ability is agent performance within a declared framework. Structured observations and mechanical shortcuts reduce the cost of reading pixels and navigating repetitive menus. Those tools are part of the task. Results do not establish unaided visual gameplay or model ability independent of the interface.

2. Methods

2.1. Tasks and starting states

The categories were navigation, battle, operations, preparation, and objectives. Operations measured manual menu use. Preparation measured choices made before battle. Objectives combined gameplay skills to complete a quest. Every model received the same registered save and task limits for each starting condition. The Supporting Information lists task descriptions and actual budgets. The companion analysis records starting-state and evaluation-specification checksums.

Curated battle checkpoints specify the player's team, health, moves, and inventory. Preparation tasks accept a constrained plan and grade the resulting team with a fixed battle controller. The model does not play those grading battles. Preparation scores depend on both the plan and reference controller. They do not estimate the win probability of an optimal player.

Revised forest, cave, boulder, and blocked-route navigation fixtures have trainers already defeated and random wild encounters disabled. This prevents incidental battles from becoming the main navigation challenge. Original navigation attempts remain archived, and their charges remain in accounting. Other tasks retain their declared game mechanics.

2.2. Agent interface and budgets

Gameplay observations included the current screenshot and structured player information: local map, party, moves, bag, visible dialogue, and recent actions. Agents could request further information through inspection. Private evaluator flags and future random outcomes were excluded. Menu-operation tasks disabled shortcuts

that would solve the tested operation. Strategy tasks allowed mechanical commands for items, moves, and switching while leaving those choices to the model.

The pass used current-image presentation and requested low reasoning effort. Recent context was bounded by 8 turns and a configured 12,000 token threshold. Context rebuilding avoided paid summarization calls. These settings do not establish identical hidden computation across providers. Recorded model identifiers appear in Table 5.

Decision limits were the primary task budget, with higher token limits as spending backstops. Decisions, controller actions, and emulator frames are different quantities. A run reaching its decision limit is a finished scored failure, not a pending job. Reported tokens include input and output usage, including cached input where the provider reports it. They are not converted to subscription dollars.

2.3. Scoring and aggregation

Gameplay tasks used their recorded completion score. Preparation used the reference controller's win rate after the model's submitted changes. Registered variants were averaged within score groups. Groups were averaged within categories, then categories received equal overall weight. The 48 task entries form 47 score groups because a related operational variant shares a group. Multiple starts do not make one battle count more than another.

Infrastructure errors were tracked separately from model outcomes. Repaired attempts replaced affected comparison cells, while original artifacts and charges remained archived. There were 0 unresolved evaluation errors in the initial matrix. This does not prove that every interface behavior was correct. Rejected responses can occur inside a completed attempt, so the audit reports them separately.

2.4. Failure review and repeats

The audit counted rejected response decisions and inspected selected traces. A rejected response is an interface symptom, not automatically a parser bug. Repeated movement within one map suggests stalled navigation, but the trace alone cannot exclude a controller defect.

The repeat check selected 3 tasks after examining initial results: forest navigation, the underleveled Lorelei battle, and move preparation. Models received additional attempts from the same saves with unchanged task budgets and the latest repaired runtime. The plan registered 36 additional attempts before execution. These repeats measure variability on selected conditions. They neither sample all game randomness nor estimate reliability across the full suite. Repeats are reported separately rather than giving selected tasks extra leaderboard weight.

3. Results

3.1. Overall and category scores

Opus obtained the highest initial overall score, 73.7%, followed by Sol at 66.0% (Table 1). Category results showed different strengths. Sol scored highest in battle, while Sonnet scored highest in preparation (Table 2). These are descriptive comparisons from recorded starts, not statistical claims of superiority.

Model	Score (%)	Tokens (M)	Finished
Opus	73.7	34.91	50/50
Sol	66.0	43.89	50/50
Terra	64.0	46.23	50/50
Sonnet	62.2	43.09	50/50
Haiku	60.7	62.51	50/50
Luna	52.0	47.67	50/50

Table 1: Initial focused-pass results. Overall score is the equal-category average in percent. Token totals cover selected current attempts, excluding archived attempts. Finished includes successful and unsuccessful model outcomes.

Model	Nav.	Battle	Menus	Prep.	Quests
Opus	66.7	75.0	100.0	60.0	66.7
Sol	33.3	83.3	100.0	58.0	55.6
Terra	33.3	72.2	100.0	70.0	44.4
Sonnet	33.3	50.0	100.0	72.0	55.6
Haiku	33.3	63.9	100.0	62.0	44.4
Luna	33.3	47.2	85.0	61.3	33.3

Table 2: Category scores in percent. Preparation uses the fixed controller's win rate after a model submission. Other categories use task completion scores. Categories receive equal overall weight.

Difficulty was uneven. All models achieved full scores on 20 tasks, while all scored zero on 6 tasks. Easy tasks provided capability coverage but little separation. Universal failures need further feasibility and trace review before being attributed to model limits. Neither extreme alone proves that a task should be removed.

3.2. Failure evidence

The audit found 51 attempts reaching the decision limit and 26 ending in a failed task condition. Across 23,702 recorded gameplay decisions, 124 were rejected for providing no controller action or valid history request. Errors occurred in 11 runs, with 112 concentrated in Saffron Gym navigation. This cluster merits focused review rather than attributing every failed task to formatting.

Selected traces showed distinct problems. Sol and Opus remained on Mt. Moon's entrance floor when their decision limits expired. Opus's boulder trace recorded an unsuccessful pushing route and a plan to reset the room. Its species-capture attempt reached a Pikachu battle but had not caught it at the limit. These observations establish unfinished progress without assigning sole responsibility to the model or framework.

The all-zero league-preparation result also concealed partial progress. Its reference controller reached late opponents, including the Champion, before losing. This is a limit of the terminal win-rate score, not evidence that the submitted teams made no progress.

The Lorelei comparison exposed a tactical difference. Sol used Amnesia repeatedly before attacking and won. Opus used Hypnosis, switched early to Pikachu, and later lost. Neither attempt recorded a response-format error. Strategy and resource use are plausible contributors, but a single battle cannot separate strategy from favorable randomness. The local leaderboard links both source replays.

3.3. Usage and repairs

The initial pass reported 289,195,307 tokens, including 10,910,493 from archived attempts. A further 6,273,189 tokens were reserved for interrupted calls with missing usage. This reserve is not measured consumption. Selected current attempts accounted for 278,284,814 tokens.

Recorded repairs included battle-free navigation fixtures, provider timeout and response diagnostics, and lossless encoding for an oversized memory inspection. The latter continued the affected run from its checkpoint with cumulative usage. Routine repairs preserved completed comparisons. The pass nevertheless spans recorded backend revisions, limiting causal claims about small ranking gaps.

3.4. Matched repeat check

At this report snapshot, 36 of 36 additional attempts were complete, using 26,241,010 reported tokens. The table labels unfinished attempts explicitly (Table 3). Across 18 selected model-task pairs, 10 had a completed repeat with a score different from its initial attempt. The full pattern is reported rather than replacing an unfavorable initial result with a better repeat.

Model	Task	Initial (%)	Repeat 2 (%)	Repeat 3 (%)
Haiku	Lorelei	0	0	100
Haiku	Forest	100	100	100
Haiku	Move prep.	60	60	60
Opus	Lorelei	0	0	100
Opus	Forest	100	100	100
Opus	Move prep.	0	100	100
Sonnet	Lorelei	0	0	100
Sonnet	Forest	0	0	0
Sonnet	Move prep.	100	100	100
Luna	Lorelei	100	100	0
Luna	Forest	0	0	0
Luna	Move prep.	100	100	100
Sol	Lorelei	100	100	0
Sol	Forest	0	100	100
Sol	Move prep.	100	100	60
Terra	Lorelei	100	0	0
Terra	Forest	0	100	100
Terra	Move prep.	100	100	100

Table 3: Matched repeat scores in percent. Initial scores are from the main pass. Repeats use the same saves and task budgets. Preparation scores are controller win rates. These selected repeats do not change main leaderboard weights.

3.5. Sensitivity to task selection

Removing each score group in turn left Opus highest and Luna lowest, but changed some middle rankings (Table 4). The calculation reused existing outcomes with the same category weighting. These ranges describe dependence on task selection. They are not confidence intervals and do not measure repeated model sampling.

Model	Best rank	Worst rank
Opus	1	1
Sol	2	3
Terra	2	4
Sonnet	4	5
Haiku	3	5
Luna	6	6

Table 4: Rank ranges after omitting each score group once. Ties share a rank. The main leaderboard is unchanged.

4. Limitations and next steps

The initial ranking supports exploratory comparison. It depends on task selection, budgets, observations, tools, and the preparation controller. Equal category weights are a design choice. Most tasks initially had one attempt per model and registered start. Subscription transports may differ in hidden reasoning, caching, latency, and model-version behavior. Token totals describe resource use under these conditions, not universal cost efficiency.

Follow-up diagnostics found that compact model messages omitted task-completion status and visible-object movement details even when the full trace recorded them. The message builder has been repaired, but the frozen repair trials and Long pilot did not use that correction. Their outcomes therefore do not measure the corrected feedback path. The next useful comparison should verify that path before another broad pass. Full-game scoring remains a separate PokeBench Long evaluation rather than another category here.

5. Availability

The local leaderboard provides task goals, outcomes, budgets, usage, and replay links for gameplay attempts. Earlier comparisons remain accessible in its archive. The analysis contains summaries and checksums rather than ROMs, emulator states, private model text, or account records. Source generators reproduce the page and report tables. This revision remains local. The earlier manuscript is preserved alongside the existing Paper Scaffold toolchain[1].

Bibliography

(1) Garrett, P. T. *paper-scaffold: provenance-checked Typst manuscripts.* https://doi.org/10.5281/zenodo.22905759.

6. Supporting Information

6.1. Recorded model identifiers

Table 5 lists provider identifiers saved with these runs. Leaderboard display names are shorthand for these identifiers.

Display name	Recorded provider identifier
Opus	claude-opus-5-5
Sol	gpt-5.6-sol
Terra	gpt-5.6-terra
Sonnet	claude-sonnet-5-5
Haiku	claude-haiku-5-5
Luna	gpt-5.6-luna

Table 5: Provider identifiers recorded for models in the focused pass.

6.2. Task inventory and budgets

Navigation appears in Table 6, battle in Table 7, manual operations in Table 8, preparation in Table 9, and objectives in Table 10.

These tables list tasks represented in the focused pass, not every proposed or historical task. Token limits are expressed in millions per attempt. Preparation uses a constrained submission followed by offline battle grading.

Task	Decisions	Tokens (M)
Reach Vermilion while the Saffron gate is closed	250	7.5
Solve Victory Road 1F and climb to 2F	300	9
Traverse Mt. Moon and reach Cerulean City	300	9
Cross Viridian Forest and reach Pewter City	200	6
Collect the Moon Stone in Rocket Hideout B2F	250	7.5
Reach Sabrina's room through Saffron Gym	200	6

Table 6: Navigation tasks and budgets.

Task	Decisions	Tokens (M)
Bruno Machamp resource tradeoff	80	2.4
Lorelei adaptation after a missed Hypnosis	60	1.8
Become Champion with Snorlax, Graveler, Charizard, Vileplume, Parasect, and Weezing	350	10.5
Become Champion with Pidgeot, Charizard, Lapras, Vileplume, Raticate, and Beedrill	400	12
Agatha Gengar recovery	60	1.8
Brock with Squirtle and Pidgey	50	1.5
Champion preserve Jolteon	100	3
Lorelei and Bruno PP gauntlet	120	3.6
Champion Blue starter-dependent lineup	100	3
Lorelei tactical v2 Poliwrath team	80	2.4
Lorelei recovery with registered bad-matchup team	60	1.8
Route 1 Rattata battle	25	0.75

Table 7: Battle tasks and budgets.

Task	Decisions	Tokens (M)
Free a bag slot and exchange the Bike Voucher	100	3
Switch from Charmander to party slot 3 Pikachu	40	1.2
Buy exactly ten Poke Balls at Viridian Poke Mart	100	3
Use one Antidote on party slot 3 Nidoran♂	40	1.2
Deposit and withdraw party slot 2 Pikachu	100	3
Use one Potion on party slot 2 Pikachu	40	1.2
Complete an inventory and party checklist	130	3.9
Move party slot 3 Pidgey to the lead	50	1.5
Use one Ether on Squirtle Water Gun	50	1.5
Teach TM34 Bide to Pikachu over Growl	70	2.1
Withdraw one Potion and use it on Pikachu	90	2.7

Table 8: Manual menu tasks and budgets.

Task	Decisions	Tokens (M)
Lorelei roster assembly from storage	1	0.05
Lance damaged team rescue	1	0.05
Brock lead selection with Squirtle and Pidgey	1	0.05
Kadabra TM choice for Lorelei then Bruno	1	0.05
Lorelei and Bruno Elixer allocation	1	0.05
Brock Pokémon Center recovery	1	0.05
Elite Four resource preparation	1	0.05
TM24 Thunderbolt allocation for Lorelei	1	0.05
Misty shopping with Pikachu and Bulbasaur	1	0.05
Champion Blue lineup preparation	1	0.05

Table 9: Preparation tasks and budgets.

Task	Decisions	Tokens (M)
Catch Abra on Route 24, train it, then defeat Misty	650	19.5
Catch level 30 Snorlax with 12 Great Balls	100	3
Find and catch a wild Pikachu from Viridian City	200	6
Obtain a Bicycle through the Fan Club quest	200	6
Help Bill and receive the S.S. Ticket	150	4.5
Collect Oak's Parcel in Viridian and return it	300	9
Rescue Mr. Fuji and obtain the Poke Flute	250	7.5
Clear Rocket Hideout and defeat Giovanni for the Silph Scope	600	18
Choose any starter in Oak's Lab	100	3

Table 10: Quest and combined-objective tasks and budgets.

6.3. Reproduction

The snapshot is `analysis/inputs/focused-v060.json`. It records aggregate scores, selected attempts, starting-state checksums, budgets, usage, audit counts, and repeat outcomes. Replay paths are relative to the local preview site. They do not imply that private game assets are publicly distributable.

Run `scripts/export_focused_analysis.py` with the initial directory as `--run`, the local website as `--web`, and the repeat directory as `--repeats`. The `--paper-input` option refreshes this report's input. From the manuscript directory, run `just assets`, `just paper`, and `just verify`. Numeric prose and tables are generated through the existing Paper Scaffold.

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